# 3D Vision - Lab 2

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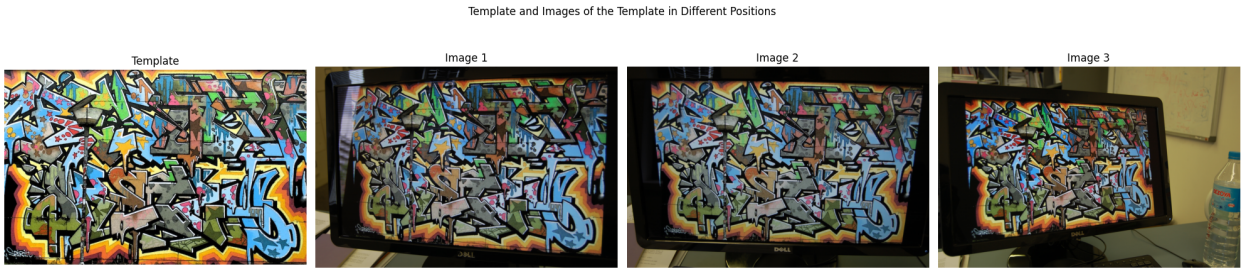
|---|---|---|
| Student Names | | |
| Student IDs | | |
| Group | | |

*Note: **Resizing answer boxes will be penalized**. Ensure your response stays within the provided borders to ensure compatibility with Gradescope.*

# 1. Calibration with a planar pattern




## 1.1 Compute Homographies Between Template and Images

You have implemented the apply_H function to apply a homography to an image.

| Paste the figure for keypoint matches between the template and the different images. |
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| • Explain how these keypoint matches were obtained.<br>• Explain the arguments and the values passed to the Ransac_DLT_homography(…) function<br>• Do the keypoint matches look right in your opinion? Are there any outliers? Where and why? |
| |

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## 1.2 Compute the Image of the Absolute Conic

| Paste the code implementation of get_v_ij_T(H, i, j) function |
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| • Explain the reasoning behind the get_v_ij_T() function. What does it represent?<br>• How many images are required to compute the IAC? Why?<br>• What would you do if you had fewer images? |
| |

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## 1.3 Compute the camera calibration matrix from the IAC

| Paste the code implementation for finding K with Cholesky. | Paste the estimated matrix K. |
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| | |

- From your computed K, identify every parameter (skew, …).
- Explain what each of those values tells you about the camera's geometry.
- Which of those entries should vary with the image's width and height? Compare your numbers to width/2 and height/2, and to a typical pixel-scale focal length.
- Are all values in a sensible range? If anything looks strange, suggest a possible source of error.

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- Paste the code for computing the field of view.
- Explain the maths behind it. (add a scheme/diagram for the explanation if necessary)
- Which are the results? Do they make sense?

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## 1.4 Compute camera position and orientation

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| Paste the code for computing the rotation and translation of each camera. |
| |
| Explain the procedure to find R, t and P with the code above in your own words. |
| |
| In which units is the translation vector **t** expressed? |
| |

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### 1.4.2 Camera-pattern geometry

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| **Camera poses relative to the calibration pattern**<br>Paste 3 screenshots from different points of view of the 3D scene (include the legend!) |
| |
| **Static camera and estimated pattern poses**<br>Paste 3 screenshots from different points of view of the 3D scene (include the legend!) |
| |
| • What is the relation between the two situations?<br>• Do the results make sense to you for camera poses relative to the calibration pattern?<br>• Explain why.<br>• Do the results make sense to you for the static camera and estimated pattern poses?<br>• Explain why. |
| |

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## 1.5 Augmented reality

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| Paste the code to project the vertices onto the image plane |
| # TODO: Project vertices onto the image plane using the projection matrix<br>… |
| Explain it with your own words. |
| |
| Paste the figure for “Augmented views: cube overlaid on the planar pattern”. |
| |
| • Comment the results obtained.<br>• Do these results make sense to you? Why? |
| |

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## 1.6 Building your own example

| Paste the Augmented views: cube overlaid on the planar pattern for your own example. |
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| |
| • Comment the results obtained.<br>• Do these results make sense to you? Why?<br>• Did you encounter any issues? |
| |

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## 1.7 Reducing the number of images

- How many degrees of freedom has the omega vector?
- Is it reasonable to assume skew = 0? Why?
- How many equations will we have if we include the assumption for skew = 0?
- What is the minimum number of views you need to calibrate the camera in this case?

- Is it reasonable to assume $(x_0, y_0)$ = (width/2, height/2)? Why?
- How many equations will we have if we include the assumption $(x_0, y_0)$ = (width/2, height/2)?
- What is the minimum number of views you need to calibrate the camera in this case?

- Write all the equations that we have for **V** matrix, and their assumptions.

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| Augmented **image 2** calibrated with **3 images** | Augmented **image 2** calibrated with **1 image** |
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| | |

- Compare the results. Are there any differences? Are they expected? Why?

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## 1.8 Augmented Reality Video

Paste 4 frames of the augmented reality video.
(Add frames properly augmented but add a frame where the reconstruction was not perfect too.)

- Did it work? Why? Or why not?
- Are there any frames where the augmentation was not perfect? Why?
- Is there any jittering? Why?

# AI usage reflection (Specific answers to this Lab)

We want you to use AI responsibly and reflect about its usage in an **honest** manner.

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| • Which AI tools have you used? (ChatGPT / Copilot / Gemini, embedded in Visual Studio, Cursor, etc)<br>• For which tasks have you used these tools? (Debugging, writing code, understanding concepts) |
| |
| • What did the tool succeed on (in this Lab)?<br>• What did the tool fail in (in this Lab)?? |
| |
| • Have you changed how you use these tools compared to the previous lab? Explain it. |
| |

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Provide s**creenshots of the prompts** you used and the resulting output from the tool:

- **Successful Example**: One instance where the tool significantly assisted you. Justify its success and usefulness.

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- **Failed Example**: One instance where the tool did not perform as expected. Explain the reason for the failure.

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# Self-awareness

- Provide an overall explanation of the lab. Describe the goals of the lab clearly. State the problem you were trying to solve or explore in this lab.

- What have you learned? Did the lab help you visualise and understand concepts?

- Which parts of the notebook did you succeed in? Describe the sections where you felt confident and explain why you think they were successful.

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| • Which parts of the notebook did you fail to solve? Be honest about the areas where you faced difficulties. What challenges or issues did you encounter that you couldn't resolve? How would you approach these issues in the future? |
| |
| • Are there any concepts that are not completely clear yet? I.e., what question would you least like to be asked about? |
| |

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# Extras

Add extra results below. If needed, you can use multiple pages for this section.